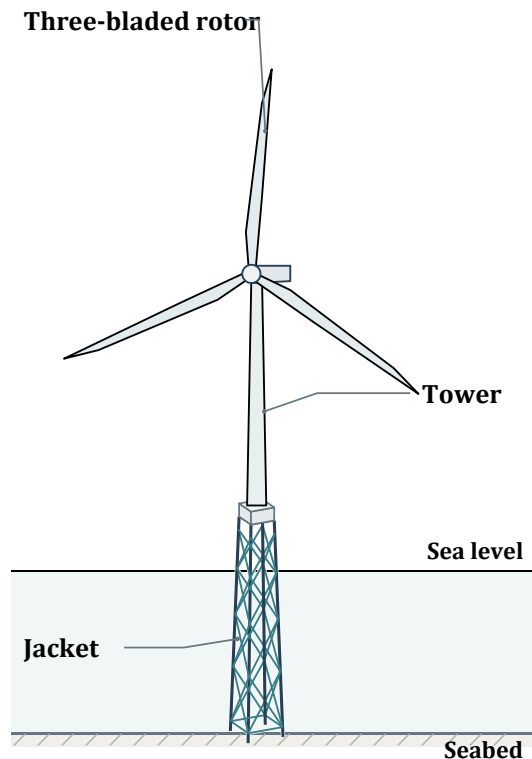
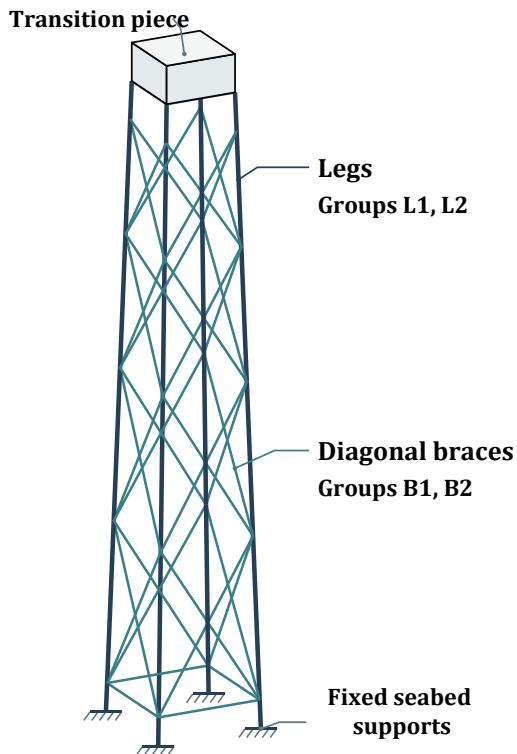


(a) Wind turbine and jacket



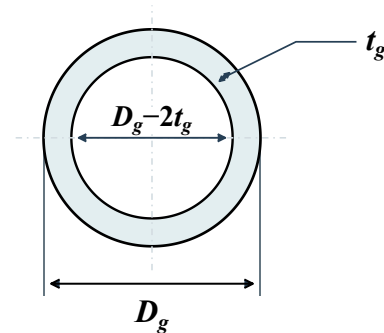
Bottom-fixed four-legged jacket

(b) Jacket parameterization



Node locations and connectivity unchanged

(c) Section and design variables



Section properties defined by D_g and t_g

Leg section groups Brace section groups

$L1: (D_{L1}, t_{L1})$ $B1: (D_{B1}, t_{B1})$

$L2: (D_{L2}, t_{L2})$ $B2: (D_{B2}, t_{B2})$

4 section groups · 8 design variables

Members in each group share D_g and t_g

$$0 < 2t_g < D_g$$